

Design, Assembly and Working Modes of Spherical 3-RRR Coaxial Parallel Robot

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Abstract—Design, configuration and operation modes of spherical 3-RRR coaxial parallel robot were performed. A mathematical analysis of the robot was made, which consists of inverse and forward kinematic using geometric and numerical methods respectively, using MATLAB environment. Based on the above, eight possible operation and configuration modes were derived. The proposed approach demonstrates that a unique solution for the specific configuration and operation mode of a coaxial robot can be calculated. Moreover, mechanical design of the robot prototype was performed in the SolidWorks design program presented in detail for experimental verification of the performed numerical and simulation analyses. Results obtained from this work can be implemented in the development of real time orientation control systems for coaxial robots. As future works, an analysis of singularities, performance and dynamic will be carried out.

Index Terms—Kinematics, Parallel Robots, model, Configuration, Operation, Mechanical Design.

I. INTRODUCTION

In recent years parallel robotic are frequently claimed to offer several advantages over their serial counterparts, such as their ability to support high loads and reach high velocities without stress, high dynamic capacities and high accuracy [1]. Thus, they are interesting for applications where these properties are needed, such as flight simulators [2] and high-speed machines. Recently, new applications have used such mechanisms to build humanoid robots [3], or medical and rehabilitation robots [4, 5]. However, they are subject to other constraints because of possible collisions of the links with each other [6, 7]. Indeed, this type of robot only can produce a limited torsion motion with a prescribed tilt angle and an unlimited torsion is necessary in some common material processing such as milling. The coaxial input described in [8] can achieve unlimited torsion, whereas its unique structure introduces a complex input mechanism.

A review of the literature shows the parallel spherical robots were studied in a variety of problems related to kinematics and performance. Some authors develop the inverse kinematics of this type of robot, where the orientation of the end-effector is determined by Euler's angles [9]. A related solution is based on the directional cosines of each joint axis, as functions of the variables of the actuated joints [10]. Likewise, Because of the architecture of this type of robot, the forward kinematics

is calculated from a closed system of equations [11], with combinations where a maximum of eight solutions of the non-linear of equations system is allowed. Solutions are derived from the roots of an eighth-order polynomial equation, known as the robot characteristic equation [12]. Methods such as Ostrowski's and Newton's homotopy are alternative solutions to this number of non-linear equations [13]. Parallel spherical robots may be configured considering the limited ranges of rotational movement, as active spherical joints. This criteria is known as a spherical 3-RRR parallel robot with coaxial input [14].

Future work focuses mainly on modeling and kinematic and dynamic analyses of this robot design. The problem of workspace analysis with respect to a spherical robotic wrist is initially approached in [15]. The wrist is designed according to a modular configuration approach and is capable of unlimited rolling rotation. Afterwards the computation of the forward and inverse kinematic model of a spherical parallel robot 3-RRR for a teleoperation solution was described by [16], real-time application environment was the objective. Particularly, several approaches for robot design optimization based on They have been extensively studied on many others aspects, such as manipulability [17], dexterity [18], and stiffness [19].

In this paper, we present a mechanical design of spherical 3RRR coaxial parallel robot with coaxial actuators according to given parameters, resulting from mathematical analysis. The kinematic performance of the proposed robot is studied and the eight operation and configuration modes are visualized for the optimum design in the further study. This work is organized as follows: Section II, kinematic Analysis and the eight operation and configuration modes are visualized for the optimum design in the further study. Section III, shows mechanical design, configuration with coaxial actuators in accordance with the parameters given and mathematical analysis considerations. In section IV, experimental validation, kinematic analysis approaches coaxial robot CAD. Finally, Section V summarizes the main conclusions and future work of this research.

II. KINEMATIC ANALYSIS AND OPERATION AND CONFIGURATION MODES

Fig. 1, illustrates the geometric description of Spherical 3-RRR Parallel Robot with coaxial actuators (3-DOF), with the respective nomenclature for kinematic modelling. This

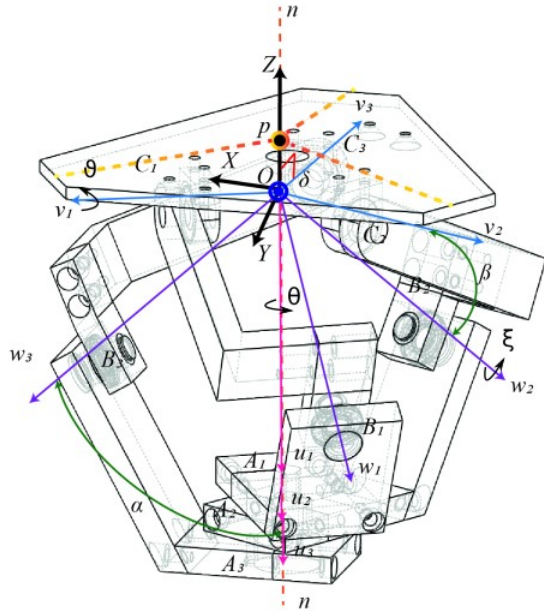


Fig. 1: Geometric Description of Spherical 3-RRR Coaxial Parallel robot [6].

manipulator consists of a rigid base, with three independent and identical kinematic chains ($i = 1, 2, 3$), attached to the end-effector. The unit vectors u_i have constant values, indicating the rotational axes of A_i joints, associated to fixed joints. On the other hand, the unit vectors w_i and v_i , have variable values indicating the rotary axes of the B_i and C_i joints corresponding to free or passive joints, respectively. The angles α and β are the dimensions of proximal and distal links; δ and γ define base geometry and end-effector respectively, where the point, P , provides the desired position of the platform [6]. On each kinematic chain, there is a fixed drive revolute joint whose rotation is associated with the angle θ_i , and two free revolute joints connecting to the proximal and distal links, respectively, and the distal link to the end-effector. The rotation of the two free joints are represented by angles ξ_i and ϑ_i respectively

A. Eight Operation Modes of the Coaxial Robot Model

The obtaining of the eight possible solutions can be observed by computing the Euler angles. In Fig. 1, the kinematic chains for each leg of the mechanism are shown. These can be written by means of rotation matrices between the fixed base of the mechanism and the end-effector.

It is possible to observe that for each chain, the transformation matrix can be expressed through Euler angles ZYX and an extra rotation around z - axes, as follows:

$$R_1 = R_z(\theta_1) \cdot R_y(\xi_1) \cdot R_x(\vartheta_1) \quad (1)$$

$$R_2 = R_z(\theta_2) \cdot R_y(\xi_2) \cdot R_x(\vartheta_2) \cdot R_z(\eta_2) \quad (2)$$

$$R_3 = R_z(\theta_3) \cdot R_y(\xi_3) \cdot R_x(\vartheta_3) \cdot R_z(-\eta_2) \quad (3)$$

where (1), (2) and (3) describe the matrix of transformation of the leg 1, 2 and 3 respectively. By writing the rotation

matrix (R_I) of each leg in a generic form as show in (4), it is possible to observe how this matrix intrinsically depends on the $\sin(\xi_i)$.

$$R_I = \quad (4)$$

$$\begin{bmatrix} C\xi_i C\theta_i & C\theta_i S\xi_i S\vartheta_i - C\vartheta_i S\theta_i & C\theta_i S\xi_i C\vartheta_i + S\vartheta_i S\theta_i \\ C\xi_i C\theta_i & S\theta_i S\xi_i S\vartheta_i + C\vartheta_i C\theta_i & C\theta_i S\xi_i S\vartheta_i - S\vartheta_i C\theta_i \\ -S\xi_i & C\xi_i S\vartheta_i & C\theta_i C\xi_i \end{bmatrix}$$

Therefore, based on this consideration, 4 possible values for $\sin(\xi_i)$ can be identified:

- 1) Case 1: $\sin(\xi_i) \neq \pm 1$

Given (4), it be calculated respectively the value of θ_i , ξ_i and ϑ_i for $i = 1, 2, 3$, obtaining the two sets of angles for the two solutions [20], as indicated in (5) and (6).

$$\begin{cases} \theta_{i1} = \text{atan2}\left(\frac{R_{21}}{\cos\xi_i}, \frac{R_{11}}{\cos\xi_i}\right) \\ \xi_{i1} = \text{asin}(R_{31}) \\ \vartheta_{i1} = \text{atan2}\left(\frac{R_{32}}{\cos\xi_i}, \frac{R_{33}}{\cos\xi_i}\right) \end{cases} \quad (5)$$

$$\begin{cases} \theta_{i2} = \theta_{i1} + \pi \\ \xi_{i2} = \pi - \xi_{i1} \\ \vartheta_{i2} = \vartheta_{i1} + \pi \end{cases} \quad (6)$$

- 2) Case 2: $\sin(\xi_i) = \pm 1$

The literal terms of the generic matrix R change depending on the value of the $\sin(\xi_i)$ and $\cos(\xi_i)$.

Thus, the generic matrix begins as expressed by (7) and (8) :

$$\begin{aligned} \bullet \sin(\xi_i) &= 1 \\ \xi_i &= \pi/2 \end{aligned} \quad (7)$$

$$\begin{aligned} \bullet \sin(\xi_i) &= -1 \\ \xi_i &= 3\pi/2 \end{aligned} \quad (8)$$

It is possible to solve the inverse kinematics problem in the above cases using the atan2 function, obtaining (9) and (10).

$$\begin{aligned} \bullet \sin(\xi_i) &= 1 \\ \vartheta_i &= \text{atan2}(R_{12}, -R_{22}) + \theta_i \end{aligned} \quad (9)$$

$$\begin{aligned} \bullet \sin(\xi_i) &= -1 \\ \vartheta_i &= \text{atan2}(-R'_{12}, R'_{22}) - \theta_i \end{aligned} \quad (10)$$

The eight operation modes are obtained using Algorithm 1, in the only considers as input the orientation of the end-effector ($\theta_i, \vartheta_i, \xi_i$).

B. Forward Kinematic

Forward kinematics of the parallel spherical 3-RRR robot, base on geometric constraints, describes the orientation of the end-effector defined between the unit vectors v_i and Euler angles (φ, σ, ϕ), given the position vector of the actuated joints (θ_i).

Algorithm 1: Eight operation modes of the coaxial Robot model

Input: φ, σ, ϕ
Output: $\theta_i, \vartheta_i, \xi_i, i=1,2,3$

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1  $Rot_{EE} = R_x(\pi/6)R_y(\pi/2)R_z(\pi/4)$ 
2 if  $Rot_{EE}(3,1) < 1$  then
3   % case 1: if  $R_{31} \neq \pm 1$ 
4   if  $Rot_{EE}(3,1) > 1$  then
5      $\theta_i = \text{atan2}(\text{rotm}(2,1), \text{rotm}(1,1))$ 
6      $\xi_i = \text{asin}(-\text{rotm}(3,1))$ 
7      $\varphi_i = \text{atan2}(\text{rotm}(3,2), \text{rotm}(3,3))$ 
8   else
9     % case 2: if  $R_{31} = -1$ 
10     $\theta_i = -\text{atan2}(-\text{rotm}(2,3), \text{rotm}(2,2))$ 
11     $\xi_i = \pi/2$ 
12     $\varphi_i = 0$ 
13 else
14   % case 2: if  $R_{31} = +1$ 
15    $\theta_i = \text{atan2}(-\text{rotm}(2,3), \text{rotm}(2,2))$ 
16    $\xi_i = -\pi/2$ 
17    $\varphi_i = 0$ 
18 return  $\theta_i, \vartheta_i, \xi_i$ 

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Based on the geometric constraints of the robot, it is possible to obtain a system of 3 independent equations, from each kinematic chain, as show in (11) [21].

$$\begin{cases} w_i \cdot v_i = C\beta \\ v_i \cdot v_j = C\zeta \\ ||v_i|| = 1 \end{cases} \quad (11)$$

C. Eight Configuration Modes of the Coaxial Robot Model

Equation (11) consists of three linear and six quadratic equations of unit vectors v_i . This system cannot be solved analytically, so numerical methods are used [21]. As a result of this, it is necessary to define a vector of initial values, in which the signs of the values are essential and depend on the initial framework. Modifying the signs of the initial vector values, v_i , eight possible configuration modes can be found for the spherical 3-RRR coaxial parallel robot. The availability of various solutions is guaranteed since there are two directions to place the distal link for any position of the proximal link of the same leg. Consider the case of the forward kinematics of the coaxial robot model with $\theta = [55^\circ, 90^\circ, 75^\circ]^T$. To determine the resulting orientation of the mobile platform of the coaxial robot, i.e. to compute unit vectors v_i from (11).

To obtain the unique solution of this forward kinematics problem, the signs of the x, y and z components of v_i , at the home configuration are used in the initial guess vector $x_0 = [v_{1x}, v_{1y}, v_{1z}, v_{2x}, v_{2y}, v_{2z}, v_{3x}, v_{3y}, v_{3z}]^T$. Therefore, the kinematic solution is considered to be in the same configuration mode as the initial coaxial robot configuration and will correspond to the physical prototype of the robot.

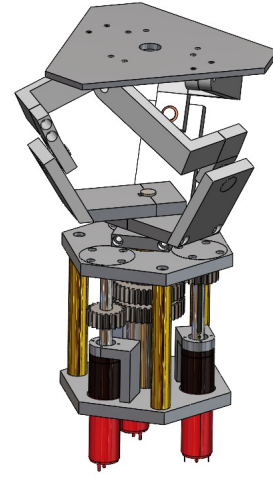


Fig. 2: 3D CAD model of the coaxial robot - Isometric View.

III. DESIGN OF PARALLEL SPHERICAL 3-RRR ROBOT

The proposed algorithms for calculating kinematics are demonstrated numerically in a coaxial robot model with the following geometrical parameters: $\alpha = \pi/3$, $\delta = \pi/2$, $\beta = \pi/2$, and $\gamma = 0$. A 3D CAD (Computer Aided Design) model of the coaxial robot configuration corresponding to these parameters has been designed in the Solidworks © software and is presented as shown in Fig. 2.

A. Design Using Three Motors

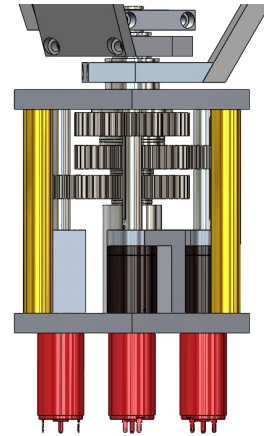


Fig. 3: Three motors are arranged in a symmetrical pattern - Frontal View.

As shown in Fig. 3, the three motors are configured in a symmetrical pattern. In the center of the base, through the bracket, the mechanism consisting of the three coaxial shafts with the respective pair of radial bearings has been allocated. At the bottom, each shaft has a built-in gear coupled with the respective motor. Therefore, the gears are connected with the motor thanks to the Bearing Bronze Bushing Plain. In the same way, a pivot connects each gear to the upper platform,

to avoid longitudinal bending due to the radial force generated during the meshing of the teeth. It is fixed with respect to the gear, while it has been coupled with a bearing to the upper platform, for the purpose of allowing the rotation of the gear with the pivot. A bearing has been inserted in the upper base, to avoid the same longitudinal curve for the shaft mechanism.

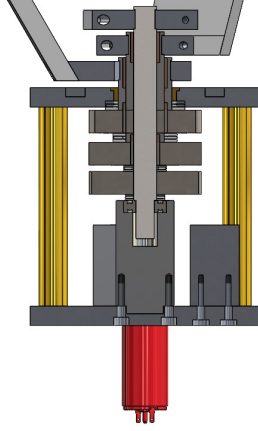


Fig. 4: Three motors are arranged in a symmetrical pattern - gear transmission.

To make the system reconfigurable, the drive system has been decoupled from the robot configuration as shows Fig. 4. This choice has complicated the design, however, it allows to change the characteristic angle of the robot and to rebuild it in the cheapest and fastest way. Moreover, the legs of the robot have been used to do the bearing arrangement of the shaft mechanism as show in Fig. 4.

Regarding the legs, the distal links and the end-effector (E.E.), they have been divided into different parts, since their shape is complex to realize in one piece and there are several orientations driven in contrast to each other, during the creation of the layer. For these reasons, it has been preferred to separate these bodies in order to save mechanical quality and characteristics such as stiffness, bending, torsion and density.

In the end, the realisation of the three independent degrees of freedom and the respective relative rotation among the legs, arms and the end-effector it has been possible thanks to a couple of Bronze bushings, inserted in each head of these bodies and a pivot that realises this connection as presented in Fig. 5. The choice to use a couple of bearing has been made to reduced sensitivity to the misalignment of the pivots during the mounting of the robot.

IV. RESULTS AND DISCUSSION

Fig. 6 and Table I show the eight operation modes, which are obtained from the orientation described in the (12).

$$\mathbf{R}_{EE} = \mathbf{R}_x(\pi/6)\mathbf{R}_y(\pi/2)\mathbf{R}_z(\pi/4) \quad (12)$$

To get case 1, only were considered solutions with a positive sign since it limits the ranges of values $([-\pi/2, \pi/2])$ to ξ_i . On the other hand, in case 2, there is a gimbal lock between θ_i

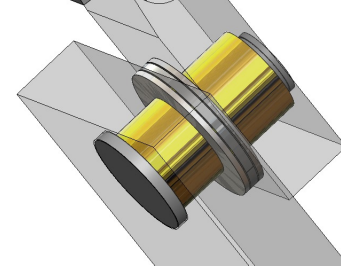


Fig. 5: Detail - end-effector and Distal Link Joint Section.

Solution	Leg 1			Leg 2			Leg 3			Possible Solution
	θ_1	ξ_1	ϑ_1	θ_2	ξ_2	ϑ_2	θ_3	ξ_3	ϑ_3	
1	90	-60	30	9	41	55	-140	13	-64	Yes
2	270	240	210	9	41	55	-140	13	-64	No
3	90	-60	30	189	139	235	-140	13	-64	No
4	270	240	210	189	139	235	-140	13	-64	No
5	90	-60	30	9	41	55	40	167	116	No
6	270	240	210	9	41	55	40	167	116	No
7	90	-60	30	189	139	235	40	167	116	No
8	90	-60	30	189	139	235	40	167	116	No

TABLE I: Angular joint position for each operation modes of the Robot.

and ϑ_i , since the axis of two of the three rotors are located in parallel, producing a loss of a degree of freedom in one of the three rotors. As a consequence, there are many solutions to (10). Therefore, to determine the equation the value of $(\theta_1, \theta_2, \theta_3)$ are assumed as 0° .

In particular, solution 1 shown in Fig. 6a, represents one possible solution, while the others are impossible, due to the physical interference between the legs. Therefore, it is possible to define from the inputs $(\theta_1, \theta_2, \theta_3)$ three fixed frames that distance between them 120° , under the following conditions:

- $120^\circ - \theta_1 - \theta_2 > 15^\circ$
- $120^\circ - \theta_2 + \theta_3 > 15^\circ$
- $120^\circ - \theta_3 + \theta_1 > 15^\circ$

where 15° is the limit angle, chosen to avoid physical interference between the legs.

Eight kinematic solutions from (11) defined by several possible assembly modes are illustrated in Fig. 7 and listed their angular joint and end-effector position in Table II and Table III, respectively. These solutions are obtained from the numerical solution in MATLAB © using the vector of initial values $x_0 = [1, 1, -1, 1, -1, 1, 1, 1, 1]^T$ described in the Section II.

Notably, in the configuration modes of the robot before being attached to the end-effector, each distal link can be aligned to the left (clockwise) or to the right (counterclockwise). Interestingly, based on the results obtained in the Fig. 7, the arrangement of all distal links rotated in the same direction, i.e., left-left-left (l-l-l) or right-right-right (r-r-r), is more appropriate as it provides symmetry. Thus, Left-left-left (l-l-l) configuration mode for the designed coaxial robot model and the prototype is selected in this work as shows Fig. 7a.

V. CONCLUSIONS AND FUTURE WORKS

In this paper the approach to calculate the 8 possible unique solutions that allow to determine configuration modes and

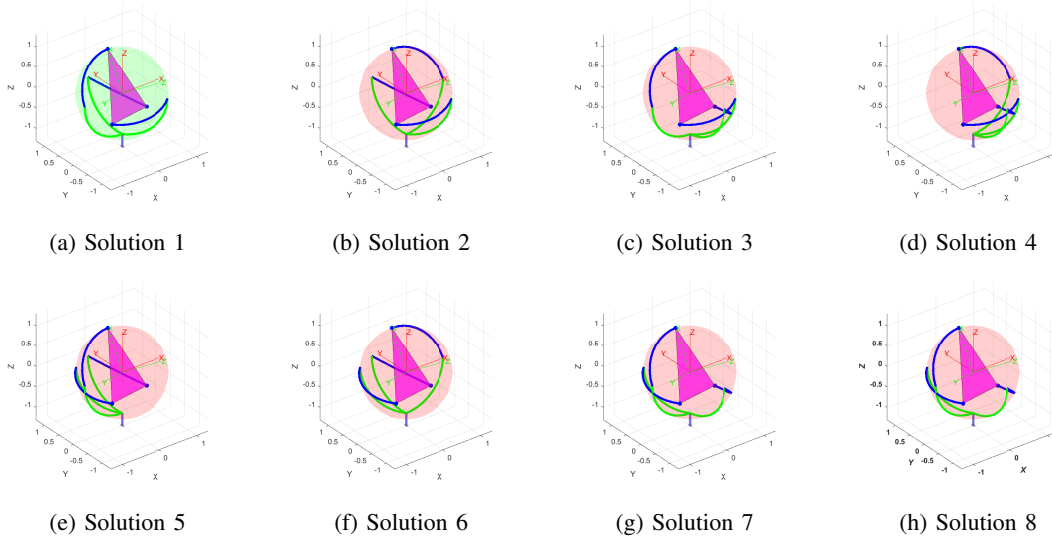


Fig. 6: Eight operation modes of the coaxial parallel Robot.

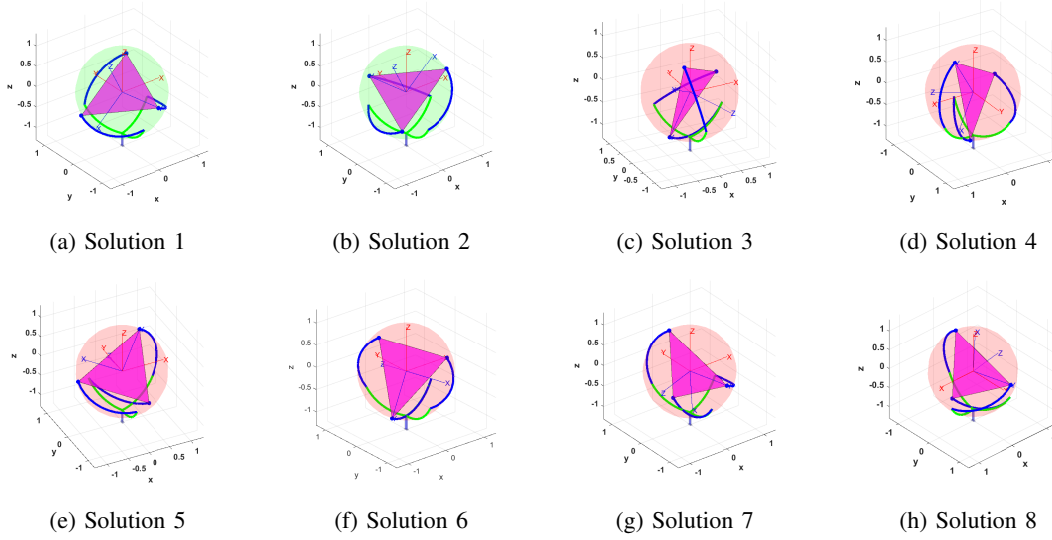


Fig. 7: Eight configuration modes of the coaxial parallel Robot

Solution	φ	σ	ϕ
1 (l-l-l)	177.9103	149.8249	12.9304
2 (l-l-r)	-104.5133	-26.7255	-141.3639
3 (l-r-l)	104.3521	-26.6448	193.6206
4 (l-r-r)	-2.1928	-94.8213	17.1031
5 (r-l-l)	182.1928	85.1787	17.1031
6 (r-l-r)	-104.3521	26.6448	-13.6206
7 (r-r-l)	104.5114	26.7272	38.6364
8 (r-r-r)	2.0668	-30.1831	12.9074

TABLE II: Eight forward end-effector angles kinematics solutions of the coaxial robot model with $\theta = [55^\circ, 90^\circ, 75^\circ]$.

operation modes of a coaxial robot with joints of revolution has been described in detail. Several positions for both modes have been analyzed allowing to define three constraints to

avoid collisions between legs at the moment of operation due to the configuration of the links. In addition, the two optimal configuration modes (left-left-left (l-l-l) or right-right (r-r-r)) have been defined for the appropriate performance of this type of manipulator and any other with different geometrical parameters. As a future work of the presented robot, an analysis of singularities, performance and dynamic and a control algorithm by combining the fuzzy approach with PID sliding mode control method will be carried out.

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Solution	v_1			v_2			v_3		
	v_{1x}	v_{1y}	v_{1z}	v_{2x}	v_{2y}	v_{2z}	v_{3x}	v_{3y}	v_{3z}
1 ($l-l-l$)	0.2413	-0.97	-0.0312	0.6087	0.6524	0.4514	-0.8506	0.3174	-0.4192
2 ($l-l-r$)	-0.4964	-0.0763	-0.8647	0.8526	0.5208	0.0428	-0.3558	-0.4451	0.8217
3 ($l-r-l$)	0.4806	0.1382	0.866	0.5119	-0.2519	-0.8213	-0.9926	0.1126	-0.0445
4 ($l-r-r$)	-0.257	0.9664	0.0033	0.1987	-0.4616	-0.8645	0.0596	-0.5044	0.8614
5 ($r-l-l$)	0.257	-0.9664	-0.0033	-0.1987	0.4616	0.8645	-0.0596	0.5044	-0.8614
6 ($r-l-r$)	-0.4806	-0.1382	-0.866	-0.5119	0.2519	0.8213	0.9926	-0.1126	0.0445
7 ($r-r-l$)	0.4964	0.0764	0.8647	-0.8526	-0.5208	-0.0428	0.3558	0.4451	-0.8218
8 ($r-r-r$)	-0.2411	0.97	0.0315	-0.6094	-0.6523	-0.4507	0.85	-0.3179	0.4201

TABLE III: Eight forward kinematics solutions of the coaxial robot model with $\theta = [55^\circ, 90^\circ, 75^\circ]$.

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